Evaluation Metrics

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Confusion Matrix

Definition: The confusion matrix shows how many predictions fall into each category of true vs. predicted labels.

		Predicted	
		Positive	Negative
Actual	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

Accuracy

Definition: Proportion of correctly predicted samples.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Example: 80 correct out of 100 predictions $\Rightarrow$ Accuracy $= 0.8$

Use: Works best for balanced datasets.

Sensitivity / Recall

Definition: Measures how well the model detects positive cases.
Sensitivity is also termed as Recall or **True Positive Rate (TPR)** which measures how well true positives are identified by a test or model. It depicts among all the positive cases how many were correctly identified as positive by the model.

$$Recall = \frac{TP}{TP + FN}$$

Example: 45 detected out of 50 actual positives $\Rightarrow$ Recall $= 0.9$
Interpretation: High recall means fewer false negatives.

Specificity

Definition: Measures how well negatives are correctly identified. Specificity is also also termed as **True Negative Rate (TNR)** which assess how well the model correctly predicts the absence of a condition and then avoids false alarms.

$$Specificity = \frac{TN}{TN + FP}$$

Example: 40 out of 50 negatives correctly classified $\Rightarrow$ 0.8
Interpretation: High specificity means fewer false positives.

Precision

Definition: Fraction of predicted positives that are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

Example: 45 true positives out of 60 predicted positives $\Rightarrow$ 0.75
Interpretation: High precision means fewer false alarms.

F1 Score

Definition: Harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Example: Precision $= 0.75$, Recall $= 0.9 \Rightarrow$ F1 $= 0.82$
Interpretation: Balances precision and recall.

ROUGE Score

Definition: Measures textual overlap (n-gram or longest common subsequence) between generated and reference summaries.

Example:
Reference: *"The cat sat on the mat."*
Prediction: *"The cat is on the mat."*

Overlapping unigrams = 5, total in reference = 6
ROUGE-1: $\frac{5}{6} = 0.83$

Note: ROUGE counts overlapping word occurrences, not just unique words.

Use: Evaluating summarization, translation, and text generation quality.

BERT Score

Definition: Measures semantic similarity using contextual embeddings.

Reference: *"A man is eating food."*
Prediction: *"A person is having a meal."*

Result: BERTScore ≈ 0.89
Use: Evaluates meaning similarity beyond word overlap.